

Teaching at the Edge: On-Device Parameter-Efficient Adaptation and What It Actually Buys

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Abstract—A companion study evaluated local inference on consumer hardware, measuring request-window GPU board energy of 0.091 Wh for 3B and 0.19 Wh for 8B on an RTX 5070, and replacing unconditional carbon, water and cost headlines with explicit accounting boundaries and limits on cloud comparisons [1]. That study treated the small-model-sufficient workload fraction f_s as fixed by parameter count. This paper tests whether parameter-efficient fine-tuning (PEFT) moves it, on hardware the household already owns. We model sixteen 2026-era adaptation methods, three consumer device classes, and the memory, time, energy and money cost of an adaptation run; we check the memory model against a publicly documented 27B-parameter training run to within -0.6% and compare throughput against the same run’s wall clock. We state five hypotheses and three counter-claims. Our central result is negative with respect to the obvious coverage hypothesis: at the companion study’s acceptance threshold $\alpha = 0.70$, the modelled adaptation changes coverage by $+0.0$ pp - not at all, because every adaptable task class already cleared the bar. This invariance is structural conditional on the declared class ratios and workload weights. A public-safe held-out evaluation across three models, five synthetic task classes, 20 evaluation tasks per class and three adapter-training seeds measures task accuracy from 86.7% to 97.8%, a $+11.1$ pp point estimate with a model-cluster bootstrap interval of 5.0–18.3 pp. The exact cluster sign-flip test is $p = 0.250$ with only three model clusters, so this is evidence for the declared synthetic benchmark, not a strong population-level significance claim. The earlier $+2/+6/+14$ points and the derived $88.4\% \rightarrow 93.9\%$ figure remain exploratory scenario inputs for routing sensitivity, not the measured quality result. The measured benchmark is not a sampled user workload, a human quality study, or an independent device replication. Coverage responds only above $\alpha \approx 0.80$ in the exploratory scenario, and that threshold depends on the uplift assumed. A single public recipe reports a 27B adaptation run; applying its reported 130 W sustained package power to 82.1 hours implies 10.7 kWh. Against an all-cloud counterfactual that is repaid in 143 days of ordinary use, but charging it only against the routing the adapter actually changes - the correct attribution - it is never repaid at $\alpha = 0.70$, because it moves no query, and takes 1.8 to 3.3 years where it does. We report both bounds and record H8 as accounting-dependent rather than validated. Repeating adaptation per user is scale-invariant under this model, which answers the “centralized training amortizes better” objection on its own terms. A private one-deployment report provides bounded, deployment-reported observations: five rank-32 LoRA runs are reported as moving one task from 30.8% to 92.9%, while refusal of out-of-scope requests moved from

0% to 100% and source-code leakage from 25% to 0%. The underlying prompts, transcripts, checkpoints, and task-level labels are not redistributable or independently auditable here, so these observations are not a population estimate or a replacement for the planned experiment. They delimit, rather than dispute, the literature concern that fine-tuning can degrade safety alignment. As a separate prompt-surface robustness probe, changing the surrounding instructions while reusing the public evaluation records gives 80.0% to 90.0% accuracy, or $+10.0$ pp with a model-cluster interval of $+5.0$ – $+15.0$ pp and exact $p = 0.250$. This probe is not a new semantic workload, an independent replication, or evidence that the primary effect generalises beyond the released records. A separate new-semantic holdout uses 100 fresh synthetic records, 20 per class, with new record values and task formats. Across the same three model clusters and three adapter seeds it gives 90.3% to 91.8%, or only $+1.4$ pp, with a model-cluster interval from -3.3 pp to $+7.7$ pp and exact $p = 1.000$. The model-level directions are mixed, so this result weakens a broad semantic-transfer reading of the primary uplift. It remains synthetic, automatically graded, and measured on one physical host.

Index Terms—parameter-efficient fine-tuning, LoRA, QLoRA, edge computing, sustainable AI, on-device training, privacy, small language models

I. INTRODUCTION

The companion study [1] asked whether it is feasible to run a small model at home. Measuring an NVIDIA RTX 5070 under Ollama at 499 input and 300 output tokens, it observed median request-window GPU board energy of 0.091 Wh for 3B and 0.19 Wh for 8B, while establishing that environmental comparisons against cloud baselines remain conditional on workload matching, task success, grid intensity and whole-system counterfactuals. Under an assumed small-model-sufficient workload fraction $f_s \approx 82\%$, decentralised execution trades off against cloud alternatives, but against a hyper-efficient cloud model at 0.24 Wh per prompt [2] a single consumer GPU is worse once the comparison is made on like terms — the board-rail figures above exclude host draw and supply loss, whereas the cloud figure is a comprehensive at-the-wall measurement — and on genuinely hard reasoning a small model is not a substitute at all.

Throughout, one quantity did the work. f_s is computed from per-task-class capability ratios - small-model quality divided by frontier quality - and those ratios were treated as properties

of parameter count. This paper asks what happens when they are not.

Two things changed in 2026 that make the question answerable rather than speculative. First, the open-weights tier a household can hold moved from 8B to 27B: Qwen3.8-27B [3] is dense, natively multimodal, Apache-2.0 licensed, and carries a 262K-token context. The official Ollama distribution lists an 18GB Q4_K_M build [4]; that is distribution metadata rather than a measurement of whole-system memory or runtime energy. Second, consumer unified-memory APUs put 128GB in front of the accelerator, and a public recipe documents multi-day 27B LoRA training on one of them [5]. Adapting a 27B model has therefore been demonstrated outside data-centre hardware, although the recipe remains a single community validation point.

Our contributions:

- 1) A **cost model for on-device adaptation** covering memory, wall clock, energy, carbon and money, checked against a publicly documented 27B run to -0.6% on peak memory (§III).
- 2) A **conditional negative result on the obvious coverage hypothesis**: under the declared ratios, adaptation does not increase coverage at an ordinary acceptance threshold; a separate public synthetic benchmark measures task-accuracy change without claiming a real-workload effect (§IV).
- 3) An **amortisation analysis** charging the full training footprint against the inference it displaces, under two accountings that disagree by more than an order of magnitude, and a correction to our own earlier reading of it (§V).
- 4) A **delimitation of the fine-tuning safety result**, supported by one deployment’s reported runs (§VI).
- 5) A **scale-invariance argument** against the centralized-amortisation objection (§VII), and an explicit separation of which of our results are structural and which are inherited from an assumption (§IV-E).

All parametric models and figures are deterministically recomputed from grounded equations and documented parameters; none is transcribed.

II. THE 2026 METHOD ROSTER

Sixteen methods are modelled. They divide into families that behave very differently on a memory-bound device, and the division matters more than any individual method’s reported gain.

Unless a sentence below is a definition of the method or a library compatibility fact, the stated gain, speed, or forgetting property is an author-reported result from the cited paper’s own tasks and protocol. None of these sixteen methods is re-run in this study. We use the reported ranges only to bound a design sensitivity; they are not a matched ranking of methods on the private deployment workload.

Every method below is a variant of low-rank adaptation [6]; [7] surveys the family, and [8] documents which of them sit

one keyword argument away in the library a household would actually use.

Free. *rsLoRA* [9] scales adapters by $\alpha/\sqrt{r}$ instead of α/r . This is one keyword argument with an outsized effect: under the original scaling, raising the rank shrinks the effective update, so higher ranks stop helping - which is why so much practice settled on $r \in \{8, 16\}$ and concluded that rank does not matter. It does; the scaling was hiding it. *LoRA+* [10] gives the two factors different learning rates; its paper reports an infinite-width result and faster convergence under its evaluated settings. *NEFTune* [11] adds embedding noise and reports its largest gains on conversational quality specifically. None of the three costs additional persistent adapter memory in our accounting; their training-time overhead is not measured here.

Initialisation. *PiSSA* [12] starts the adapter in the principal singular subspace of W rather than at noise-and-zero; *OLoRA* [13] uses QR-based orthonormal initialisation; *EVA* [14] runs a few batches of the *actual* dataset and allocates rank where that dataset needs it; *CorDA* [15] can build the update in directions the base model does not use for general knowledge. *LoftQ* [16] initialises the adapter to absorb quantization error, so training does not begin by repairing the damage quantization did. Their one-time initialization costs are outside our measured adapter transport model; no recurring cost is claimed here.

Quantization. *QLoRA* [17] combines NF4, double quantization of the block constants (≈ 0.37 bits per parameter), and paged optimizer states. It is one route by which a 27B model can fit on consumer hardware, but the fit and training-time tradeoff depend on context, optimizer, checkpoint, and runtime. This study does not re-measure its wall clock or memory profile.

Reparameterisation. *DoRA* [18] splits the update into magnitude and direction; its paper reports larger gains at low rank in its evaluated settings. That is a hypothesis for the edge, not an edge measurement. *AdaLoRA* [19] reallocates a fixed rank budget across layers during training. *VeRA* [20] and *LoRA-XS* [21] go the other way, cutting trainable parameters by over $100\times$ in a reported 7B comparison, with the quality tradeoff depending on task and rank; §VIII explains why that trade is interesting for a distributed product even when it is uninteresting for a benchmark.

Wrong shape. *GaLore* [22] reports a higher continued-pretraining ceiling than LoRA in its evaluated settings, but it updates all weights and therefore produces a new model rather than a mergeable adapter. For a system whose distribution unit is a small personal delta, we treat that shape as outside the adapter-only transport design regardless of quality. We include it to make the criterion explicit: the artifact’s *shape* is a first-order design constraint, not an implementation detail.

Fig. 1 places all sixteen on cost against reported gain.

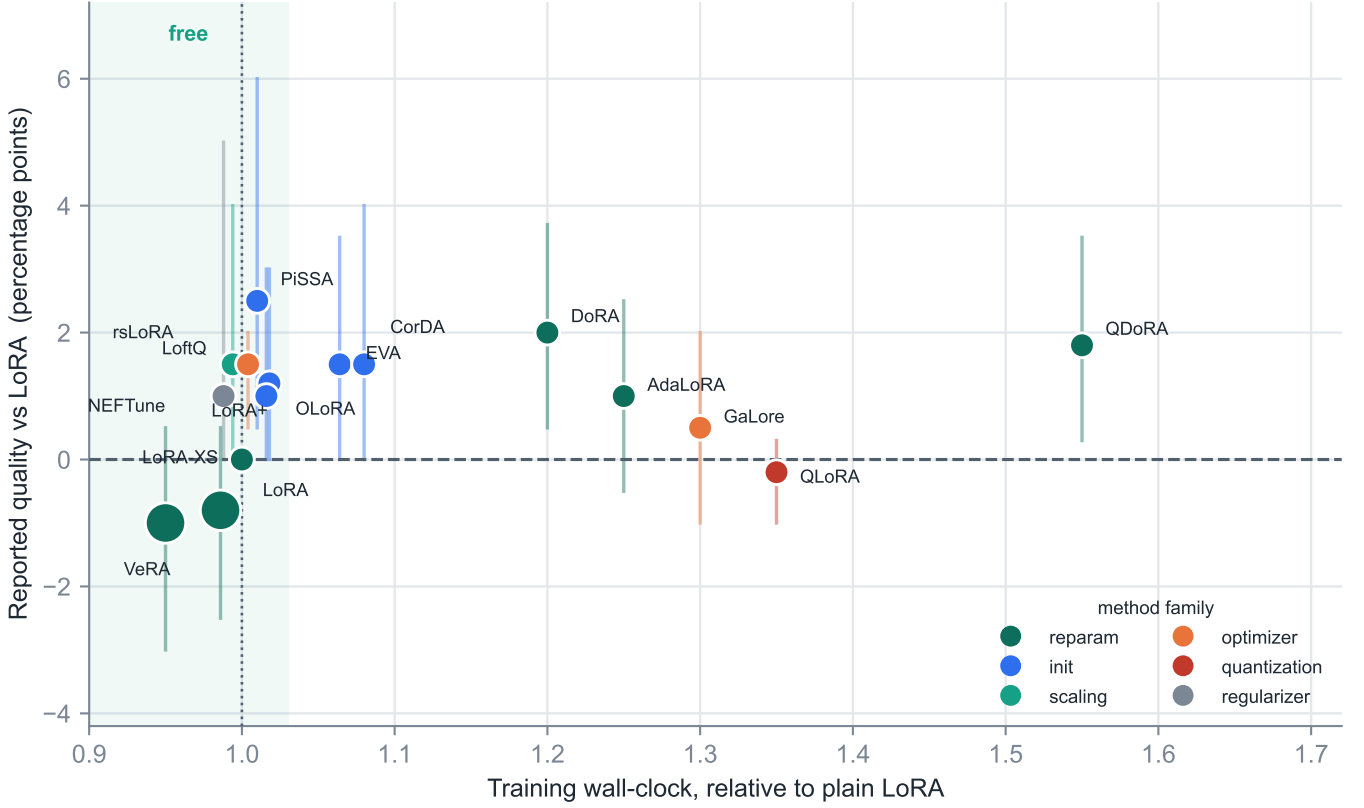


Fig. 1. The 2026 roster. Vertical bars span each method’s reported range against a LoRA baseline; the dot is the modal reported gain; large dots shrink the adapter by more than $20\times$. Three methods sit on the 1.0 line: reported gains at no wall-clock cost. Ranges are the methods’ own authors’ benchmarks on their own tasks and are used here only to bound sensitivity.

III. COST MODEL AND VALIDATION

A. Memory

Peak accelerator memory for an adapter job is

$$M = \underbrace{Pb_w}_{\text{weights}} + \underbrace{n_a(b_a + b_g + b_o)}_{\text{adapter, grads, optimizer}} + A + L + \varepsilon, \quad (1)$$

where P is the parameter count, b_w the bytes per stored weight (2 for bf16, 4.127/8 for NF4 with double quantization), n_a the trainable adapter parameters, A the activation term, L the logit term = $BSV \cdot 4$ bytes, and ε an 8% allocator margin. With gradient checkpointing, $A = L_{\text{layers}} BSH \cdot 2 + BSH \cdot 34 \cdot 2$.

The logit term is easily overlooked and is not small: at $S = 8192$ over a 152K vocabulary it is ≈ 5 GB by itself, computed in fp32 for every position.

B. Validation

One 27B run is publicly documented in enough detail to check against: Qwen3.5-27B [23], LoRA $r = 128$, $\alpha = 256$, bf16 weights, paged 8-bit AdamW, sequence 8192, batch 1, gradient accumulation 4, on a 128 GB unified-memory APU, reported at ≈ 80 GB peak reserved [5]. Our model predicts **79.5 GB**, a -0.6% residual, decomposing as 54.0 weights, 6.4 adapter, gradients and optimizer combined, 8.2 activations, 5.0 logits, 5.9 allocator.

C. Time and energy

Adapter training costs $\approx 6N$ FLOPs per token with checkpointing: $2N$ forward, $2N$ backward to activations (still required to reach earlier layers even with frozen weights), $2N$ for the recomputed forward; the adapter’s own gradient is $O(n_a)$ and rounds away. The same publicly documented run gives 448 steps $\times$ 11 minutes at 32,768 tokens per step, from which the achieved throughput is **8.04 TFLOP/s** - a model-FLOPs utilisation of 0.136 against the part’s ≈ 59 TFLOP/s bf16 peak, and 49.6 tokens per second. At 130 W sustained package power the full run consumes **10.7 kWh** over 82.1 hours.

D. Device class dominates

Table I states the paper’s least intuitive result. The *faster* consumer device cannot adapt the larger model. A discrete GPU with four times the memory bandwidth is driven down to rank 32 at sequence 1024 before a 27B job fits, while a unified-memory APU runs it at rank 128 and sequence 8192 with 20 GB to spare. Bandwidth decides how quickly a run finishes; addressable memory decides whether it can start.

This is why the companion study, which modelled only the discrete-GPU class, had no way to reach the question this paper asks.

TABLE I
PEAK TRAINING MEMORY, $r=128$, SEQUENCE 8192

	dGPU 24 GB	APU 128 GB	NPU
Usable memory	19 GB	102 GB	13 GB
Bandwidth	1008 GB/s	256 GB/s	120 GB/s
Training power	430 W	130 W	–
27B bf16	no	yes	no
27B 4-bit	$r=32, S=1024$	yes	no
8B bf16	yes	yes	no

IV. DOES ADAPTATION MOVE f_s ?

A. What can and cannot be adapted

An adapter learns the shape of a distribution it was shown. It can learn how a person phrases things, which of their files matter, what a product’s screens are called, and when to refuse. It cannot teach a small model competition mathematics, because the training set contains no such capability to distil: the ceiling on hard reasoning is the base model’s depth, and PEFT does not raise it. We therefore mark four of the five workload classes adaptable and hold the hard-reasoning class fixed - which means the companion study’s central negative finding survives this paper unchanged.

The existing +2, +6 and +14 capability-ratio points (low, central, high) are exploratory sensitivity inputs, not observed effects. They are retained only to show how the routing arithmetic would respond to a bounded uplift, with the pessimistic reading of Biderman *et al.* [24] as context rather than as a measured effect size. Ratios are capped at 0.99: an adapter may close the gap to a frontier model on a narrow task, but no claim is made that a rank-32 update exceeds one. A new public multi-model, multi-task evaluation replaces this scenario for the narrower estimand of held-out synthetic task accuracy. It does not replace the real-workload sample, the human grading plan, or the independent-device replication still needed for a product-level claim.

B. The result

At $\alpha = 0.70$, f_s is unchanged: 82% $\rightarrow$ 82%, a gain of +0.0 pp. Every adaptable class already cleared the bar without adaptation, so there is nothing for the uplift to recruit.

H9 as stated remains only partially supported. The public benchmark measures held-out task accuracy, not the real-workload fraction f_s . Across the declared 3 models and 300 model-task observations, accuracy rises from 86.7% to 97.8%, a measured change of +11.1 pp (cluster bootstrap 5.0–18.3 pp). The exact cluster sign-flip test is $p = 0.250$: the direction is consistent across the three model clusters, but three clusters do not support a strong population-level significance claim. The gain is concentrated in hard-reasoning tasks (+38.9 pp); extraction and retrieval questions are already at 100.0% and 100.0% in this benchmark. This is a measured synthetic-task result, not evidence that a deployed assistant improves by the same amount. The class-level effects are secondary: the hard-reasoning cluster test has raw $p = 0.250$, and the five-test

Bonferroni-adjusted value is $p = 1.000$; no task-class effect is confirmatory in this three-cluster study.

C. Prompt-surface robustness

The primary benchmark’s disjoint task identifiers do not by themselves show that an adapter learned transferable semantics. We therefore ran a separate probe on the same public evaluation records after changing the outer instruction, class-specific wording, and labels surrounding the answer choices. The probe assigns new task identifiers and uses the same three model clusters, five classes, and three adapter-training seeds, but it deliberately does not claim a new workload sample.

On this changed prompt surface, base accuracy is 80.0%, adapted accuracy is 90.0%, and the frontier reference is 95.0%. The adapted-minus-base change is +10.0 pp with a model-cluster bootstrap interval of +5.0–+15.0 pp and exact two-sided cluster sign-flip $p = 0.250$. The direction and scale are similar to the primary result, which weakens the simplest objection that the effect is caused only by the exact outer wording. It does not answer the stronger objections: the underlying semantic records and answer labels are reused, only one paraphrase is tested, contamination is not probed, and the run remains on one physical host. The holdout is therefore a robustness check, not an independent replication or a product-quality estimate.

D. New-semantic holdout

The prompt-surface probe above reuses semantic records, so we also generated a separate holdout with new synthetic records, new domain values, and a different reasoning format. It contains 20 tasks per class across extraction, retrieval, summary, simple code, and hard reasoning. The released task manifest has no primary evaluation task identifiers or source-task links, and the runner records that primary prompts and answers are not copied. The same three model clusters and three saved adapter-training seeds are scored with the same blinded greedy four-choice grader.

On this new semantic holdout, base accuracy is 90.3%, adapted accuracy is 91.8%, and the frontier reference is 95.0%. The adapted-minus-base change is +1.4 pp, with a model-cluster bootstrap interval from –3.3 pp to +7.7 pp and exact two-sided cluster sign-flip $p = 1.000$. The model-level changes are +7.7 pp, –3.3 pp, and 0.0 pp for the 0.5B, 3B, and VL 3B clusters respectively. The mixed directions and interval spanning zero are counterevidence against treating the primary +11.1 pp as a broad semantic-transfer effect. This is still a synthetic accuracy check on one host; it does not test contamination, untouched general ability, human task success, sampled user work, or independent-device reproducibility.

Under the central exploratory scenario, mean capability on the served workload would rise from 88.4% to 93.9% of frontier, a derived gain of +5.6 pp that closes 48% of the remaining gap. Those values remain arithmetic consequences of the scenario, not observations. In that same scenario, coverage responds only above $\alpha \approx 0.80$, where the unadapted model begins failing classes the adapted one still passes: at $\alpha = 0.80$

coverage goes $70\% \rightarrow 82\%$, and at $\alpha = 0.90$, $48\% \rightarrow 70\%$ (Fig. 2).

E. Which half of this result is evidence

Those two numbers are not the same kind of claim, and conflating them is the easiest way to misread this paper.

The *coverage* result is algebraically invariant conditional on the declared class partition, base ratios and workload weights. At $\alpha = 0.70$ every adaptable class already sits above the threshold, so no positive uplift can recruit additional workload: we obtain $f_s = 82\%$ for every uplift tested from 0 to +30 points. It does not establish that a sampled user workload has those ratios or weights; a task corpus with classes near the threshold could change the result.

The *measured synthetic-task* magnitude is a finding only for that predeclared benchmark. It uses deterministic public templates, four-choice greedy grading, one physical host and three model clusters. The exact cluster-level $p = 0.250$ is not persuasive evidence of a general effect with only three clusters, even though every model’s point estimate is positive. It also says nothing about human preference, refusal quality, long-horizon task completion, or distribution shift.

The +5.6pp quality figure is different. It is the assumed uplift propagated through the workload weights and the 0.99 cap, arithmetically identical to the +6-point central scenario in §IV; the derived “48% of the gap closed” inherits that input directly. The measured benchmark now replaces the scenario for the narrow accuracy estimand, but not for this workload-weighted quality figure or for the coverage claim.

The knee location is also scenario-dependent. At the low end of the uplift range the $\alpha = 0.80$ knee disappears and coverage first moves at $\alpha = 0.90$; at the central and high ends it sits at 0.80. “Coverage responds only to a demanding bar” is robust; the specific threshold is not, which is why Fig. 2 plots the full band rather than the central line alone.

The correct claim is narrower than the one we set out to test. The declared ratios imply that adaptation would not let a local model attempt more of this work at $\alpha = 0.70$. The public benchmark suggests that adaptation can improve correctness on these synthetic tasks, but it does not establish that users experience the same gain. A sampled real workload, multiple physical device classes, and blinded human or task-specific grading remain necessary before promoting this to a product-level quality claim.

V. AMORTISATION

An adapter is not free. We first charge the full training run against the per-query saving of serving locally rather than in the cloud, with two accounting choices made deliberately against the hypothesis: training is charged at *full* device power rather than marginal, and savings are measured against a *hyperscale* cloud query, which is the cloud’s best case. §V-A then asks whether that saving is the adapter’s to claim, and concludes that at an ordinary acceptance threshold it is not.

Every case repays inside a year on this accounting, and three of the four inside a month (Table II). The 27B capability run

TABLE II
ADAPTATION BREAK-EVEN AT 50 QUERIES/DAY

Run	Hours	kWh	Queries	Days
Persona 8B, dGPU	0.5	0.21	140	3
Persona 8B, APU	3.3	0.43	288	6
Support 7B VLM, APU	12.4	1.61	1,073	21
Capability 27B, APU	82.1	10.67	7,127	143

- four days of continuous training - is repaid after about five months of ordinary use.

A. Repaid to whom? A correction

Table II charges the training run against the cloud-minus-edge saving on *every* query the adapted model then serves. Read against §IV, that premise does not survive. If adaptation moves no query across the routing boundary, those queries ran locally before the adapter existed and their saving accrues whether or not it exists; crediting it to the adapter counts a benefit the base model already delivered.

Charging only the coverage the adapter actually moves gives a different and less flattering picture:

Threshold	Coverage moved	Attributable saving	Payback
$\alpha = 0.60$	+0%	0 Wh/query	never
$\alpha = 0.70$	+0%	0 Wh/query	never
$\alpha = 0.80$	+12%	0.18 Wh/query	3.3 years
$\alpha = 0.90$	+22%	0.33 Wh/query	1.8 years

At the study’s own default threshold the adapter has no energy payback at all. It displaces nothing, because there was nothing left to displace; its training cost is a net addition, bought for quality rather than for footprint. Payback appears only in the regime where adaptation moves the routing boundary, and even there it is measured in years rather than months.

The two accountings are not competing arithmetics for one quantity - they answer different questions. Table II correctly answers “how long does a local deployment take to repay an adaptation run against an all-cloud counterfactual,” which is the right frame for a user who would have abandoned the local model and gone to the cloud had its answers not improved: someone whose *revealed* acceptance threshold sits above the coverage knee even though their stated one does not. That is a behavioural assumption, we state it as one, and we report both bounds rather than choose between them.

H8 is therefore recorded as **partial and accounting-dependent**, not validated. The two accountings answer different questions, and which one applies is a property of the deployment rather than of the adapter.

The companion study’s caveat applies on top of both readings. Payback scales inversely with the cloud baseline being displaced; against a hyper-efficient 0.24 Wh prompt rather than a 1.8 Wh one, every figure above lengthens by roughly an order of magnitude.

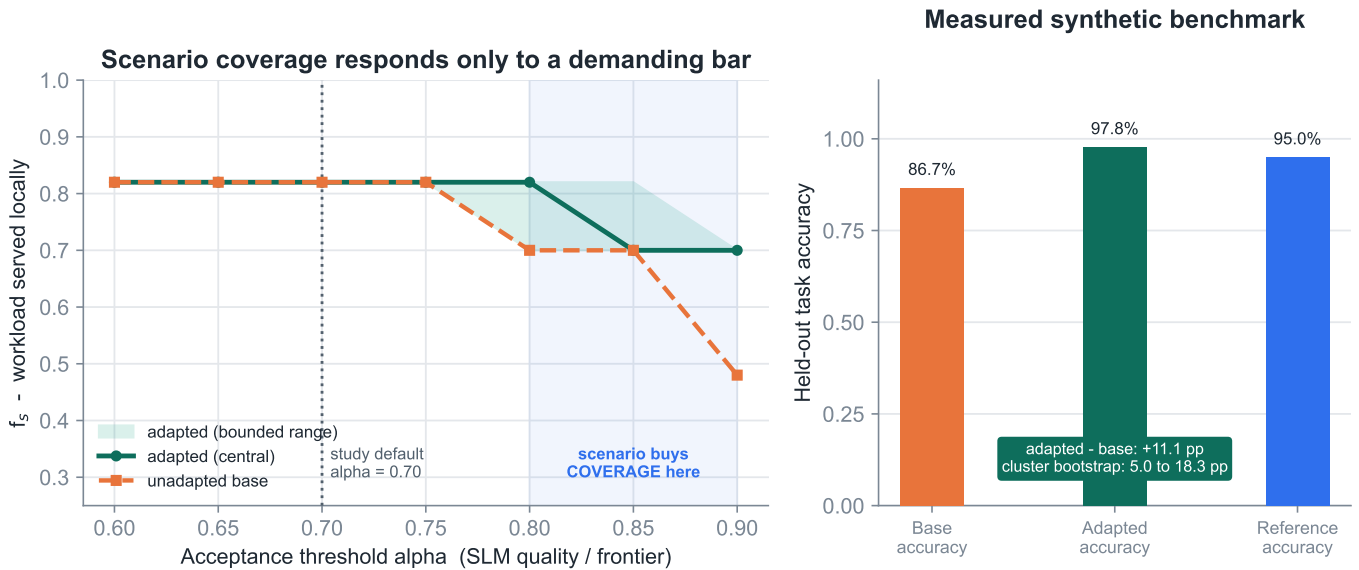


Fig. 2. The two evidence boundaries in the adaptation analysis. Left: coverage against the acceptance threshold under the inherited workload-ratio scenario, with and without the assumed uplift. Right: measured held-out accuracy on the public synthetic benchmark for the base, adapted and reference models. The right panel is not a human study, a sampled workload, or an independent device replication.

B. Money, and the surcharge that compounds

The training bill is the smaller half of the comparison. Hosted fine-tuning charges once for training tokens, and then may charge a persistent endpoint premium. Our dated third-party price snapshot reports a $1.5\times$ tuned-endpoint multiplier for three of the four entries [25]; that is a scenario input, not a provider-wide fact, and only the cheapest entry in the snapshot has no such premium. Over one year at 50 queries per day, a locally adapted 27B costs \$1.71 of training electricity plus \$0.73 of inference electricity; the cheapest hosted equivalent costs \$7.05 of training plus \$109.50 of inference - $48\times$ the total, rising to $218\times$ against a mid-tier hosted model. A local adapter leaves the per-query cost exactly where it was.

VI. SAFETY, DELIMITED

Qi *et al.* [26] showed that fine-tuning removes safety alignment - with ten adversarial examples for under \$0.20, and measurably even with entirely benign data. Any product that hands users a trainer inherits this, and we do not dispute it.

A. Measured counter-evidence, and what it does not show

One deployment’s support model has been adapted five times, with aggregate scores retained [27], over Qwen2.5-VL bases [28]. The underlying prompts, answers, checkpoints, trainer logs, and task-level labels are not in the public release. The report also does not expose enough checkpoint lineage to independently verify the claimed base pairing. We therefore retain the table only as a deployment-reported observation, not as a reproducible experiment or a multi-model comparison. A rank-32 LoRA over a reported 1,750-sample corpus produced:

	before	after
Screen identification	30.8%	92.9%
Refuses out-of-scope code questions	0%	100%
Leaks source code	25%	0%

The capability result (E1) is a reported observation that adaptation can move a task class on this deployment. It is not an independently verified existence proof: the base checkpoint, task-level outputs, grader, and exact train/eval separation are not public, and the reported parameter pairing cannot be audited from this release.

The safety result (E2) runs opposite to [26], and the reason is instructive rather than contradictory. That corpus contains 120 refusal samples - 6.9% of it - so refusal was *in the training distribution*. Qi *et al.* describe adapters that are *silent* about safety: a gradient has no reason to preserve behaviour that nothing in the corpus rewards, so it decays. The correct generalisation is neither “fine-tuning is safe” nor “fine-tuning is unsafe” but: **fine-tuning moves whatever the corpus specifies, in whichever direction the corpus specifies, and lets everything else decay.**

For a locally-adapted assistant this is the difference between a hazard and a control surface - but only if the product treats it as one. Safety cannot be inherited by a user-trained adapter; it has to be a non-optional part of every corpus the system assembles, plus a probe that runs before an adapter is installed. That is a build requirement, and we record it as one.

A third finding (E3) was previously framed as a 3B-versus-7B model-selection comparison. We withdraw that interpretation from the public result: the base labels and checkpoint lineage are not independently auditable here. No claim about parameter count or data being the binding constraint is made until matched base and adapted manifests are released.

B. Scope

One deployment, one product’s surface area, probe sets of 12–14 screens and 8 code probes scored by an automated judge. These are deployment-reported observations, not independently reproducible existence proofs or effect sizes, and nothing in this paper treats them otherwise. Aggregate scores only were read; no prompt, answer, screenshot or user content enters the analysis.

VII. FORGETTING AND SCALE

A. C4: forgetting

The objection that narrowing a small model destroys its general ability is weaker than intuition suggests. Biderman *et al.* [24] report that LoRA learns *less* in-domain than full fine-tuning and correspondingly *forgets less* out-of-domain in their evaluated settings. We do not re-run their regularizer comparison, so the low-rank constraint is a plausible mechanism, not a product-level guarantee. Three structural defences add to it - the base weights are never modified, so a misbehaving adapter is unloaded rather than un-trained; CorDA’s knowledge-preserved mode [15] addresses the objection at initialisation; and the unadapted hard-reasoning class continues to route out regardless.

The residual risk is real and we do not model it away: a personal corpus is small, unbalanced and self-similar, which is the regime where over-fitting is most likely. The mitigation is unglamorous - a held-out split and a general-ability regression probe before installation.

B. C6: does repetition lose at scale?

The strongest form of the centralized objection is that a hyperscaler fine-tunes once, at enormous but amortised cost, while decentralized adaptation repeats the work for every user; at sufficient N , repetition must lose.

It does not, and the reason is arithmetic rather than rhetorical. Both the training term and the inference-saving term scale linearly in N , so their ratio is constant: the payback period is 12 days at 10^5 users and 12 days at 10^8 . There is no scale at which the comparison behaves differently.

The deeper answer is that the comparison is mis-specified. A central fine-tune serving everyone is only comparable if one adapter suits everyone, and the entire point of a persona adapter is that it encodes one person. The honest comparison is N local runs against N personalised runs *plus the transfer of N private corpora to a trainer* - and the second term is what locality removes.

VIII. LOCALITY AND THE SHAPE OF THE ARTIFACT

Cloud fine-tuning requires the training corpus to reach the trainer. For a persona adapter that corpus *is* the user’s message history, so the privacy question is settled before any policy is written: either the messages were uploaded or they were not. Local adaptation makes “we never received it” a property of the topology rather than a promise in a document - the same class of argument as transport-layer confidentiality, verified the same way, by what the system is unable to do.

Adapter *size* then decides whether that property survives contact with more than one device. On an 8B model at $r=32$, a conventional LoRA adapter is 174.6 MB and takes 70 seconds to cross a 20 Mbit/s link: a file transfer. VeRA is 2.8 MB and 1.1 seconds; LoRA-XS is 0.52 MB and 0.2 seconds. Both are messages. That is the difference between personal adapters that sync between a person’s own devices as a background nicety and personal adapters that require an explicit chore - and it is the reason the low-parameter methods matter here despite being uninteresting on a benchmark leaderboard.

IX. THREATS TO VALIDITY

One validation point. The memory model is checked against a single publicly documented run. It reproduces that run to -0.6% , which is evidence that the terms are right, not that the model generalises across architectures. MoE models in particular are not validated and their activation term differs.

Synthetic benchmark and residual scenario. The public multi-model evaluation replaces the $+2/+6/+14$ point range for the narrower estimand of held-out four-choice task accuracy. It uses 20 evaluation tasks in each of five synthetic classes, three adapter seeds and three model clusters on one CPU-only physical host. The $+11.1$ pp estimate has an exact cluster sign-flip $p = 0.250$; this is not enough clusters for a strong generalisation claim. Hard reasoning supplies most of the gain, while extraction and retrieval are saturated at 100% in this task set. The old range is retained only as a routing sensitivity scenario, not as an empirical effect size. The workload-weighted $88.4\% \rightarrow 93.9\%$ figure is still a *consequence* of that scenario rather than evidence for it.

The task identifiers are disjoint, but the public generator reuses a small set of class templates and the same four-choice response format across the training and evaluation splits. A supplemental prompt-surface holdout changes the surrounding wording and reports 80.0% to 90.0% ($+10.0$ pp), but it reuses the same semantic records and tests only one paraphrase. It therefore weakens the exact-template explanation without distinguishing semantic transfer from memorisation of records, answer labels, recurring class structure, or contamination. A separate new-semantic holdout now uses fresh records and formats and reports 90.3% to 91.8% ($+1.4$ pp), with a model-cluster interval from -3.3 pp to $+7.7$ pp and exact $p = 1.000$. Its model-level directions are mixed, which weakens broad transfer claims, but it remains synthetic, one-host, automatically graded, and does not include a contamination or untouched general-ability probe. Semantic transfer and product-level quality therefore remain unresolved.

The workload mix is inherited, not measured. f_s , the class weights, and the base capability ratios all come from the companion study, where they are themselves modelling choices compared with NVIDIA’s reported agentic replaceability range rather than against a task corpus of our own. A different workload composition - one weighted toward hard reasoning, or toward tasks whose base ratio sits just below α - would change both results here, and the second case would make adaptation buy coverage at the default threshold. We report the

sensitivity of the knee for exactly this reason, but we cannot bound the sensitivity to a workload we did not sample.

Energy corpus and cloud baselines. The companion study provides primary 2026 consumer GPU board measurements (0.091 Wh for 3B and 0.19 Wh for 8B on an RTX 5070), but no peer-reviewed whole-system per-query measurement covers the 2026 frontier or the 2026 open-weights tier under identical prompts. Hosted pricing, open-weights architectures and adapter evaluations are current to September 2026; fleet cloud comparisons remain literature-derived estimates.

Limited replication. §VI-A is one team’s five runs on one product, and the public benchmark uses three model clusters on one physical host. Neither can estimate a population effect, independent-device reproducibility, human quality, or generality across products.

Already-owned hardware only. Every result assumes the device exists for other reasons. The companion study establishes that embodied carbon dominates for a device provisioned specifically for inference; nothing here is an argument for buying an APU.

X. CONCLUSION

One public community recipe demonstrates parameter-efficient adaptation on already-owned consumer hardware for a 27B-class model. Under the declared cost model, it is 48–218× cheaper over one year than the hosted equivalent - chiefly because hosted personalisation surcharges every subsequent query, permanently, while a local adapter does not.

Its *energy* case is weaker than it first appears, and the weakness is instructive. Against an all-cloud counterfactual an adaptation run repays inside a year in every case modelled. Against the routing it actually changes, it repays nothing at an ordinary acceptance threshold, because at that threshold it changes no routing. An adapter is an instrument for quality, and treating it as an instrument for footprint requires a behavioural assumption that should be argued rather than assumed.

What the declared model buys is not what we expected. At an ordinary acceptance threshold it adds no coverage, conditional on the chosen ratios and workload. The public synthetic benchmark measures a +11.1 pp increase in held-out task accuracy, but the result is limited by three model clusters, one physical host, deterministic synthetic tasks and automated choice grading. The supplemental prompt-surface probe remains positive at +10.0 pp, but reuses the same records and therefore does not resolve semantic-transfer, contamination, or independent-replication counterclaims. The new-semantic holdout is only +1.4 pp, with a model-cluster interval from −3.3 pp to +7.7 pp and exact $p = 1.000$. Its mixed model directions mean that broad semantic transfer is not established by the current evidence. Coverage is bought only by users who demand frontier-comparable quality in the inherited scenario, and hard reasoning remains unreachable under the stated adaptation boundary.

The practical consequence for a system builder is the inverse of the intuition: adapt to raise the standard of what you already serve locally, not to widen what you attempt. A router that

treats an installed adapter as licence to keep more work on the device will, on this model, keep the wrong work.

The private observation we did not anticipate is the safety one. Fine-tuning is neither safe nor unsafe by construction: it can move what the corpus specifies while other behavior changes or decays. That makes the corpus, not the recipe, the safety-critical artifact in a system that lets people teach their own assistant - and it makes an explicit refusal distribution and a pre-installation probe the two things such a system should not ship without. This remains an existence observation until independent products, devices and task-specific graders reproduce it.

REPRODUCIBILITY

All analytical routines, simulation models, dataset parameters and figure-generation pipelines supporting this work form a single deterministic pipeline requiring only standard scientific libraries, and produce the parameter rosters, Monte-Carlo sweeps and sensitivity curves without relying on proprietary cloud APIs. All of it is released openly under an MIT licence [29].

One class of input is deliberately not redistributable. The deployment adapter artifacts behind §VI-A come from a deployed assistant whose training corpus and evaluation transcripts contain product and user material, and only aggregate scores were ever extracted from them [27]. Those aggregates are not redistributable, so that deployment observation cannot be independently reproduced, and H11 remains *not independently validated*. The public synthetic benchmark is a separate, reproducible replacement for the narrow task-accuracy estimand: its task manifest, aggregate task-level scores, adapter files, model revisions and runtime record are released beside this manuscript. We prefer a result the reader can see is limited over one that looks reproducible and is not. Every other number in this paper is recomputed from the parameters and measurement records described above.

The prompt-surface holdout is released separately with its task manifest, aggregate task-level scores, adapter references, model revisions and runtime record. It is a robustness probe over the same public evaluation records and must not be counted as an independent device or workload replication.

The new-semantic holdout is released separately with its fresh task manifest, aggregate task-level scores, adapter references, model revisions and runtime record. It is a distribution-shift check over synthetic records, not an independent device replication or a sampled user workload.

This is research release v0.3.0 dated 2026-09-14. Venue and DOI metadata are intentionally unassigned in this release. The public multi-model adaptation score file is present and reports a measured synthetic-task accuracy result; the +2/ + 6/ + 14 scenario is not promoted to an empirical effect size for that benchmark. Repository metadata, the citation audit, and the follow-up protocol for sampled workloads and independent device replication are released beside this manuscript.

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